A. Truth Tables and Connectives

Core vocabulary

A **proposition** has a truth value. A **valuation** assigns T or F to each atomic proposition. A **tautology** is true on every row, a **contradiction** on none, and a **contingency** on some but not all. Two expressions are **logically equivalent** when their final columns match on every valuation.

Connective meanings

p	q	$\neg p$	$p \wedge q$	$p \vee q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

$p \to q$ is false only when p is T and q is F. $p \leftrightarrow q$ is true exactly when p and q match. "p only if q" and "q is necessary for p" both mean $p \to q$.

Building an eight-row table

For atoms p, q, r, list rows in a fixed binary pattern:

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Add one column for each useful subexpression, working from the innermost connectives outward. Classify only the final column.

A.3: a complete truth table

Problem. Construct a complete eight-row truth table for $\Phi = (p \to q) \vee (\neg p \to r)$. Show useful intermediate columns and classify Φ as a tautology, contradiction, or contingency.
Worked solution. List every valuation once. Compute each implication before applying the final $\vee$:

p	q	r	$\neg p$	$p \to q$	$\neg p \to r$	Φ
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	F	T
F	F	T	T	T	T	T
F	F	F	T	T	F	T

As an independent check,

$$(p \to q) \vee (\neg p \to r) \equiv (\neg p \vee q) \vee (p \vee r).$$

Reordering exposes $\neg p \vee p$, which is always true. This checks the complete table; it does not replace a requested table.
Answer. Every entry in the final column is T, so Φ is a **tautology**.

Similar example: interpreting one row

Problem. Let $p = F$, $q = T$, and $r = F$. Evaluate $(p \to q) \wedge (\neg q \vee r)$, showing the intermediate values.
Worked solution.

$$p \to q = F \to T = T,$$
$$\neg q \vee r = F \vee F = F,$$
$$(p \to q) \wedge (\neg q \vee r) = T \wedge F = F.$$

Compute the inner pieces first and the main connective last. That dependency order should match a truth table's columns.
Answer. The proposition is false on this valuation.

Common errors

Do not read $p \to q$ as a causal claim or as $q \to p$. Do not omit rows, change row order midway, or classify from a few sample rows. In $\neg(p \wedge q)$ the negation applies to the entire parenthesized expression.

A readiness check

Can you make all eight rows; compute $\neg, \wedge, \vee, \to, \leftrightarrow$; show intermediate columns; and classify the final proposition correctly?

B–C. Translation, Quantifiers, and Negation

Conditionals in ordinary language

"p if q"	$q \to p$
"p only if q"	$p \to q$
"p iff q"	$p \leftrightarrow q$
"p unless q"	$\neg q \to p$, equivalently $p \vee q$
"p whenever q"	$q \to p$

The antecedent is sufficient; the consequent is necessary.

Similar example: conditional translation

Problem. Let e mean "the account may edit" and a mean "the account is an administrator." Translate "The account may edit only if it is an administrator" into propositional logic. Explain the direction of the arrow.
Worked solution. Editing is the claim, and administrator status is a necessary condition. If editing occurs, administrator status must hold. Thus e is the antecedent and a is the consequent. Writing $a \to e$ instead would say every administrator may edit, which the sentence does not promise.
Answer. $e \to a$.

Predicates and quantifiers

A **predicate** becomes a proposition after its variables are assigned or quantified. State the domain. Then

$$\forall x\, P(x): \ P \text{ holds for every } x,$$
$$\exists x\, P(x): \ P \text{ holds for at least one } x.$$

The order matters: $\exists x \forall y\, R(x,y)$ uses one x for every y; $\forall y \exists x\, R(x,y)$ may use a different x for each y.

Similar example: quantifiers and argument order

Problem. Both variables range over the students in a class. Let $M(x,y)$ mean "student x messaged student y." Translate each formula into English and explain how its meaning differs from the first:

$$\exists x \forall y\, M(x,y), \quad \forall y \exists x\, M(x,y), \quad \exists x \forall y\, M(y,x).$$

Worked solution. In the first formula, choose x once, before checking every y. In the second, choose a sender x after each recipient y is specified; the sender may change. In the third, the quantifier order is the same as the first but the predicate's arguments reverse sender and recipient.
Answers.
1. One student messaged every student.
2. Every student received a message from at least one student.
3. One student received a message from every student.

Check both quantifier order and predicate argument order. These statements allow a student to message themself; excluding that requires an explicit condition.

Restricting an all-things domain

If the domain is all things, use predicates to restrict types:

$$\text{Every } A \text{ is } B: \forall x(A(x) \to B(x)),$$
$$\text{Some } A \text{ is } B: \exists x(A(x) \wedge B(x)),$$
$$\text{No } A \text{ is } B: \forall x(A(x) \to \neg B(x)).$$

Use implication with a universal restriction and conjunction with an existential restriction.

C: push a negation inward

$$\neg \forall x\, P \equiv \exists x\, \neg P, \qquad \neg \exists x\, P \equiv \forall x\, \neg P.$$

Work from left to right: (1) flip the first quantifier crossed by the negation, (2) keep the remaining quantified formula in the same order, and (3) repeat. Then use De Morgan's laws and conditional elimination until each negation is immediately before a predicate.

C.1(a): a complete negation

Problem. The domain is all things. Define predicates, symbolize "All dogs have fleas," negate it with no negation before a quantifier, and translate the negation into ordinary English.
Worked solution. Let $D(x)$ mean "x is a dog" and $F(x)$ mean "x has fleas." The universal claim restricts attention to dogs with an implication:

$$\begin{aligned} &\forall x(D(x) \to F(x)); \\ &\neg \forall x(D(x) \to F(x)) \\ &\quad \equiv \exists x \neg (D(x) \to F(x)) \\ &\quad \equiv \exists x(D(x) \wedge \neg F(x)). \end{aligned}$$

The last step uses $\neg(P \to Q) \equiv P \wedge \neg Q$. A counterexample must be a dog *and* lack fleas, so the existential restriction uses conjunction.
Answer. $\exists x(D(x) \wedge \neg F(x))$: some dog does not have fleas.

C.2(b): nested negation

Problem. Let P be a predicate on two variables over declared, nonempty domains X and Y. Rewrite $\neg \forall x \exists y\, P(x,y)$ so every negation appears immediately before a predicate. Here $x \in X$ and $y \in Y$ throughout.
Worked solution.

$$\begin{aligned} \neg \forall x \exists y\, P(x,y) &\equiv \exists x \neg \exists y\, P(x,y) \\ &\equiv \exists x \forall y \neg P(x,y). \end{aligned}$$

The first step flips $\forall x$ to $\exists x$; the second flips $\exists y$ to $\forall y$. Variables remain in their original order.
Answer. $\exists x \forall y \neg P(x,y)$: some x is unrelated by P to every y.

Common errors

Do not swap $\forall$ and $\exists$ without moving a negation through them. Do not reuse a single existential witness unless the formula requires one. Define both predicate meanings and argument order.

B–C readiness

Can you translate "if," "only if," and "unless"; distinguish nested quantifier orders; diagnose reversed arguments; symbolize, negate, simplify, and translate back to English?

D. Consistency and Satisfiability

What the question asks

A collection of formulas is **consistent** or **satisfiable** when at least one truth assignment makes all formulas true. It is **inconsistent** when no such assignment exists. A satisfying assignment is a witness; a contradiction is a proof that no witness exists.

A reliable workflow

1. Define one atom for each simple statement.
2. Translate every sentence before reasoning.
3. Start with facts and implications whose antecedents are forced.
4. Propagate consequences using valid inference rules.
5. If no contradiction appears, finish a complete assignment and check every condition.
6. If a contradiction appears, cite a small subset that forces both p and $\neg p$.

Useful inference rules

$p,\ p \to q \therefore q$	Modus Ponens	(MP)
$\neg q,\ p \to q \therefore \neg p$	Modus Tollens	(MT)
$p \to q,\ q \to r \therefore p \to r$	Hypothetical Syllogism	
$p \lor q,\ \neg p \therefore q$	Disjunctive Syllogism	

From $p \leftrightarrow q$, use both $p \to q$ and $q \to p$.

D.1: inconsistency and minimality

Problem. The system is in multiuser state iff it operates normally. Normal operation implies a functioning kernel. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, it is in interrupt mode. The system is not in interrupt mode. Define atoms, translate all five conditions, decide consistency, and identify a minimal inconsistent collection if one exists.

Worked solution. Let m mean multiuser, n normal operation, k kernel functioning, and i interrupt mode. The conditions are

$$m \leftrightarrow n,\ n \to k,\ \neg k \lor i,\ \neg m \to i,\ \neg i.$$

1. $\neg k$ follows from $\neg k \lor i$ and $\neg i$ (DS).
2. $\neg n$ follows from $n \to k$ and $\neg k$ (MT).
3. $\neg m$ follows from $m \to n$ and $\neg n$ (MT).
4. i follows from $\neg m \to i$ and $\neg m$ (MP), contradicting $\neg i$.

All five conditions are needed. The following valuations satisfy the other four conditions when the indicated one is removed:

Removed condition	m	n	k	i
$m \leftrightarrow n$	T	F	F	F
$n \to k$	T	T	F	F
$\neg k \lor i$	T	T	T	F
$\neg m \to i$	F	F	F	F
$\neg i$	T	T	T	T

Answer. The conditions are inconsistent; the full five-condition collection is minimal because deleting any one makes it satisfiable.

Similar example: one satisfying witness

Problem. Whenever software is being upgraded, users cannot access the file system. If users can access it, they can save new files. If users cannot save new files, the software is not being upgraded. Define atoms, translate the conditions, and give *one* checked satisfying assignment.

Worked solution. Let u mean software upgraded, a users have access, and s users can save. The translations are

$$u \to \neg a, \qquad a \to s, \qquad \neg s \to \neg u.$$

Try $(u, a, s) = (T, F, T)$. The three formulas evaluate to

$$T \to T = T, \quad F \to T = T, \quad F \to F = T.$$

Answer. The collection is consistent, witnessed by (T, F, T). This task asks for one model. If asked to list *all* models, test all eight rows or give a complete case split; one witness would be insufficient.

Similar example: a minimal contradiction

Problem. Decide whether $p \to q$, $q \to r$, p, and $\neg r$ are consistent. Identify a minimal inconsistent subset and justify minimality.

Worked solution. From p and $p \to q$, infer q by MP. From q and $q \to r$, infer r by MP, contradicting $\neg r$. To check that every premise is needed, remove one at a time:

Removed	p	q	r
$p \to q$	T	F	F
$q \to r$	T	T	F
p	F	F	F
$\neg r$	T	T	T

Each row satisfies all remaining premises.

Answer. The four-premise set is inconsistent and minimal. A larger list containing these premises would prove inconsistency but would not itself answer a request for a minimal collection.

Listing all models

For three atoms, a truth table has eight rows. Translate each condition into a column, conjoin the condition columns, and keep exactly the rows where the conjunction is T. When reasoning by cases, state why each branch is complete.

Invalid shortcuts

Affirming the consequent ($p \to q, q \therefore p$) and denying the antecedent ($p \to q, \neg p \therefore \neg q$) are invalid. "Only if" points toward the necessary condition. A conditional with false antecedent is true.

D readiness

Can you define atoms, translate every condition, give a checked satisfying assignment or a short contradiction, list all models when asked, identify forced values, and isolate a minimal contradictory subset?

E. Equivalence Derivations with Cited Rules

Permitted rules

Each may be used in either direction:

$a \to b \equiv \neg a \lor b$	Implication	(Imp)
$\neg\neg a \equiv a$	Double Negation	(DN)
$\neg(a \lor b) \equiv \neg a \land \neg b$	De Morgan	(DM)
$\neg(a \land b) \equiv \neg a \lor \neg b$	De Morgan	(DM)
$a \lor (b \land c) \equiv (a \lor b) \land (a \lor c)$	Distributive	(Dist)
$a \land (b \lor c) \equiv (a \land b) \lor (a \land c)$	Distributive	(Dist)
$(a \circ b) \circ c \equiv a \circ (b \circ c)$	Associative	(Assoc)
$a \circ b \equiv b \circ a$	Commutative	(Comm)
$a \circ a \equiv a$	Idempotent	(Idem)

Here $\circ$ is either $\land$ or $\lor$ where appropriate. Reparenthesizing requires Associative; reordering requires Commutative.

How to plan the chain

Eliminate implications first. Compare the connective shape of the two sides. Use Distributive to factor or expand, De Morgan plus Double Negation to move a negation, and then clean up order or duplicates. Change only one justified feature per line and cite exactly one rule.

E.1: factor a common term

Problem. Show $(p \to q) \land (p \to r) \equiv p \to (q \land r)$. Cite exactly one permitted rule per step.
Worked solution. Rule abbreviations are listed above.

$$\begin{aligned}
(p \to q) \land (p \to r) &\equiv (\neg p \lor q) \land (p \to r) && \text{Imp}\\
&\equiv (\neg p \lor q) \land (\neg p \lor r) && \text{Imp}\\
&\equiv \neg p \lor (q \land r) && \text{Dist}\\
&\equiv p \to (q \land r) && \text{Imp.}
\end{aligned}$$

The common disjunct $\neg p$ is what makes the Distributive step possible.
Conclusion. The expressions are equivalent; factoring the common $\neg p$ is the key step.

E.2: De Morgan structure

Problem. Show $(p \to r) \land (q \to r) \equiv (p \lor q) \to r$. Cite exactly one permitted rule per step.
Worked solution. Rule abbreviations are listed above.

$$\begin{aligned}
(p \to r) \land (q \to r) &\equiv (\neg p \lor r) \land (q \to r) && \text{Imp}\\
&\equiv (\neg p \lor r) \land (\neg q \lor r) && \text{Imp}\\
&\equiv (r \lor \neg p) \land (\neg q \lor r) && \text{Comm}\\
&\equiv (r \lor \neg p) \land (r \lor \neg q) && \text{Comm}\\
&\equiv r \lor (\neg p \land \neg q) && \text{Dist}\\
&\equiv (\neg p \land \neg q) \lor r && \text{Comm}\\
&\equiv \neg(p \lor q) \lor r && \text{DM}\\
&\equiv (p \lor q) \to r && \text{Imp.}
\end{aligned}$$

The two Commutative steps place the common r where Distributive can factor it.
Conclusion. The expressions are equivalent. The common r is factored before applying De Morgan.

E.5: nested implication

Problem. Show $\neg p \to (q \to r) \equiv q \to (p \lor r)$. Cite exactly one permitted rule per step.
Worked solution. Rule abbreviations are listed above.

$$\begin{aligned}
\neg p \to (q \to r) &\equiv \neg\neg p \lor (q \to r) && \text{Imp}\\
&\equiv p \lor (q \to r) && \text{DN}\\
&\equiv p \lor (\neg q \lor r) && \text{Imp}\\
&\equiv (p \lor \neg q) \lor r && \text{Assoc}\\
&\equiv (\neg q \lor p) \lor r && \text{Comm}\\
&\equiv \neg q \lor (p \lor r) && \text{Assoc}\\
&\equiv q \to (p \lor r) && \text{Imp.}
\end{aligned}$$

Each line changes exactly one feature and cites exactly one permitted rule.
Conclusion. The expressions are equivalent, with one cited permitted rule at each step.

Common errors

Do not cite "algebra" or "simplify." Do not silently move parentheses or terms. Do not use a truth table when a rule-by-rule derivation is requested. Make sure every intermediate expression is equivalent to the preceding one.

Whole-Core readiness

Can you complete a truth table; translate conditional and quantified language; negate nested quantifiers; prove consistency or inconsistency; and write an equivalence chain with one named rule per step? A live form selects one problem from each section A–E and is designed for 20 minutes.